

# Project 1

## AI-Powered

### Customer Churn Analytics in SaaS Industry

**\*\*You may work on any one of the following projects, Project 1 or Project 2.**

#### Problem Statement

You are working as a Data Analyst for a SaaS subscription-based company (Software as a Service). Recently, the management team noticed an increasing customer churn rate meaning more customers are leaving the service.

While they wind up operations with your company, they also fill up an exit interview.

Your task is to analyse the reasons for churn, identify high-risk customer groups, and provide actionable insights to management.

The company has provided you with two [datasets](#):

1. A structured dataset of ~95,000 customers (demographics, subscription, tenure, support tickets, churn flag).
2. An unstructured dataset of ~3,500 customer exit interviews (text feedback).

#### Objectives

By completing this project, you will:

1. Practice real-world data cleaning on structured and unstructured data.
2. Extract churn drivers from both numeric and text data.
3. Use AI tools to classify unstructured feedback and detect customer moods.
4. Build a Power BI dashboard to communicate churn insights.
5. Write a business report with practical, data-driven recommendations.

## Tasks to Perform

### 1. Data Cleaning

- Remove duplicates, fix missing values, and standardize inconsistent labels.
- Correct numeric and categorical anomalies (e.g., invalid ages, tenure, churn flags).
- For text data, clean HTML tags, emojis, truncated responses, and null-like placeholders.

### 2. Exit Reason Classification

- Use AI tools (Claude, ChatGPT, Notebook LM) to classify interview responses into categories:
  - Price, Support, Product, Performance, Onboarding, Competitor, Value.

### 3. Mood Analysis

- Extract customer mood from interviews (e.g., Angry, Frustrated, Disappointed, Neutral, Hopeful, Positive).
- Compare mood distribution across exit categories.

### 4. Exploratory Data Analysis (EDA)

- Structured data: churn % by subscription type, tenure, region, spend, and support tickets.
- Text data: top churn reasons, mood trends, reason × mood matrix.

### 5. Dashboard Creation (Power BI)

- Create an interactive dashboard with:
  - **Overview:** churn rate, trend over time
  - **Segmentation:** churn by subscription type, tenure, region
  - **Churn Reasons:** distribution of classified exit reasons
  - **Mood Analysis:** mood distribution and reason × mood crosstab

### 6. Business Storytelling

- Create a 5-slide deck with:
  - Top churn drivers (numeric + text insights)
  - Mood patterns of departing customers
  - **3 clear recommendations** to reduce churn

# **Project 2**

## **AI-Aided Product Recommendation**

### **Analytics of E-Commerce**

#### **Context**

You are working as a Data Consultant for an E-commerce organisation. Management wants to build a product recommendation system to increase cross-sell opportunities.

Specifically, they want to understand:

- Which products are frequently purchased together
- Which combinations contribute the most revenue and margin
- How to translate these findings into actionable recommendations for product pages, marketing emails, and customer campaigns

You are provided with transaction data (100,500 rows) and a product master file and a product master catalogue for product details

#### **Objectives**

1. Clean & prepare the dataset for analysis
2. Identify Product Pairs that frequently occur together in transactions
3. Quantify Pairs by Value – measure revenue and margin contribution
4. Visualize Findings – build a Power BI dashboard for cross-sell insights
5. Tell the Story – create a concise business report with top 10 actionable recommendations

#### **Data Provided**

1. **Transactions Dataset (100k rows)**

- Columns: Transaction\_ID, Customer\_ID, Transaction\_Date, Product, Category, Quantity, Unit\_Price, Total\_Price
- 2. **Product Master Catalog**
  - Columns: SKU, Product, Category, Brand, MSRP, Unit\_Cost, Margin\_Pct, Avg\_Discount\_Pct, Launch\_Date, Is\_Discontinued

## Tasks to Perform

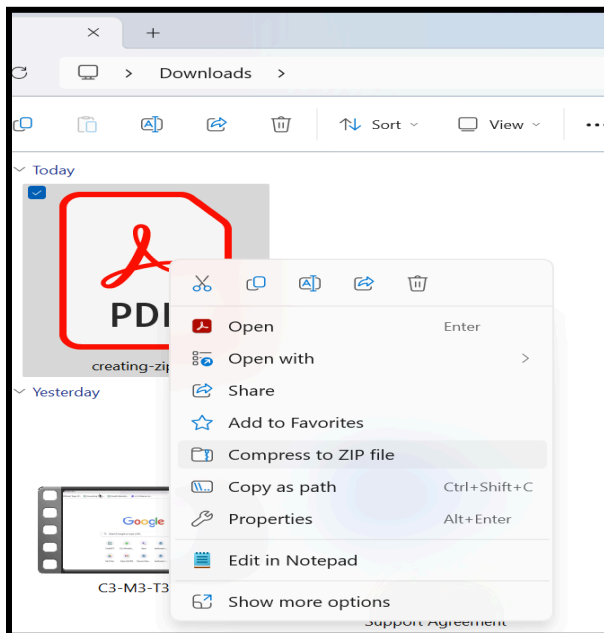
1. **Data Preparation (Excel/Python)**
  - Check for duplicates, missing values, and unrealistic prices
  - Summarize dataset (total transactions, customers, products, revenue)
2. **Co-Purchase Analysis (SQL + AI/ Excel + AI)**
  - Identify **product pairs bought together**
  - Compute metrics:
    - Pair Count (number of times pair appears)
    - Pair Revenue (total value generated by pair)
3. **Dashboard (Power BI)**
  - Dashboard Pages:
    - **Overview** → total transactions, revenue, top pairs
    - **Pair Explorer** → select Product A → see top 5 recommended Product B
    - **Revenue View** → top pairs ranked by revenue & margin
    - **Filters** → by Category, Brand, Month/Quarter
4. **Business Storytelling**
  - Create a 5-slide deck that covers:
    - Top 10 product recommendations
    - Rationale (frequency + value)
    - Exclusions (e.g., discontinued products)

## Submission Guidelines:

- Save all the files, and summary in a PDF and then convert it into a zipped (.zip) folder.
- Upload the zipped folder on your respective dashboard.

### How to ZIP a PDF file:

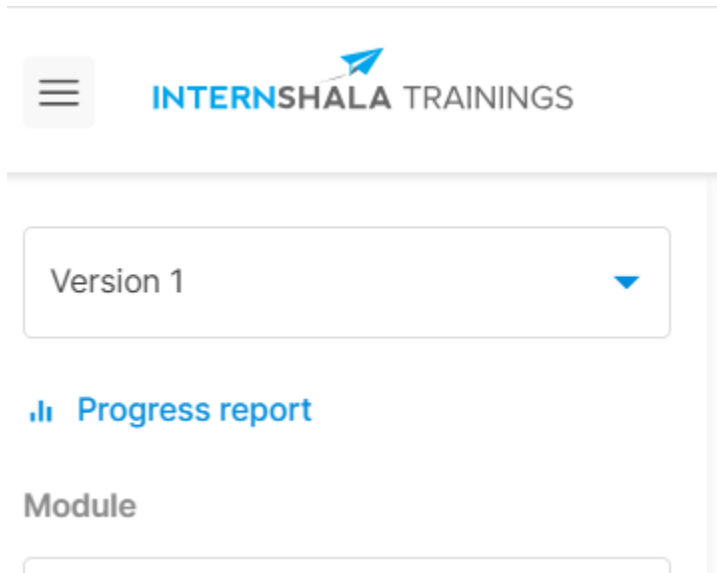
- Put all of the documents (or just one) you want to compress into a new folder.
- Right click on that folder.



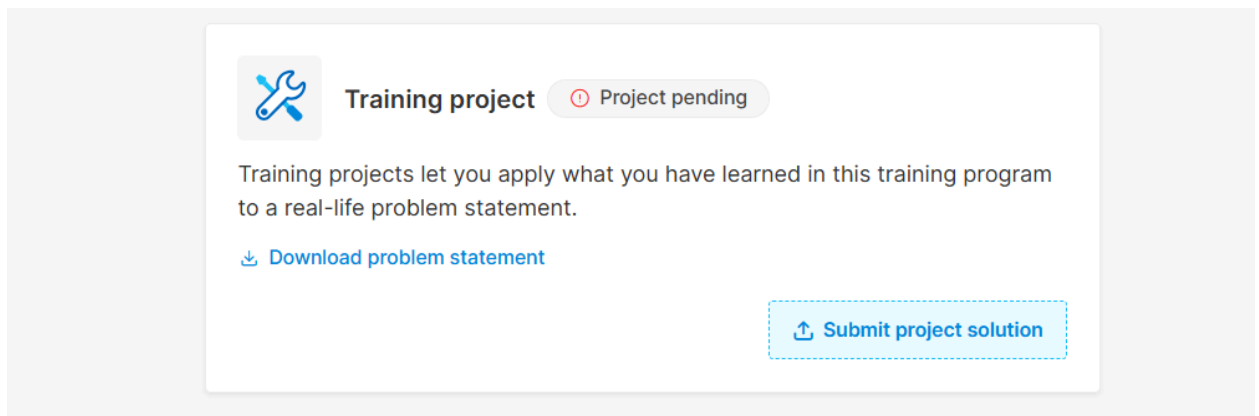
- Select the “Compress to ZIP file” option and then click “Compressed (Zipped) folder.”
- A new .ZIP file will be created that contains your document(s).

**In order to submit the projects please follow the following steps:**

1. Click on "Progress Report".



2. Next, click on the "Download Problem Statement."



3. Next, click on "Submit Project Solution".
4. Please follow the guidelines (screenshot is shared below) provided in the project to ensure correct submissions. Then, upload the zip file and click on "Upload" to

submit your work.

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### Upload here

**Instruction:**

1. You can submit your project only once. Please ensure this is the final solution that you want to submit.
2. You can upload only one .zip / .rar file (<40MB). Make one .zip / .rar file containing all assignment files in it and upload.

 [Choose file](#) No file chosen

[Upload](#)

 **Certificate**